The shared record

Above: ARCHITECTURE.md §4.7 (outside the box) and §5 (how the blocks share data).

Talks to:

Program	What it does here
the desk	writes the facts it fetches, and the copies of the registers and of the market history
the research worker	writes the O'Neil facts
qp	reads registers, market states and facts, as of a day
the journal	reads what was known at a trade's entry

The backups dump and verify it every night.

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1. What it is

One database (Postgres) for what more than one program, or more than one tool, needs:

It holds	Why it is shared
facts about the world, each with its time and source	short interest, news, the CANSLIM letters and the rest are the same whichever tool asks. Stored once, with the instant they became known, they give every reader the same answer "as of" any instant
a copy of the market history: the market states and the hot sectors	the desk keeps the history in its own file; the copy lets qp read the market as of any day
the O'Neil facts	written by the research worker, read by the desk, qp and the journal
a copy of every tool's records: what it found each morning (R1), how those stocks moved (R3), its shortlist, its screeners and their versions	the copy keeps the research record safe if the box is lost between two nightly backups, and lets qp read any tool's history as of any day

It is not the record of a tool's results. Each tool's own file on the box is (FILES.md). So the desk keeps working when the shared record is down: the tools work from their own files and fetch their own facts, and the copies catch up.

What it never does:

- holds a secret;
- is written by a page directly;
- has two writers for one fact family or one table.

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2. The diagram



3. Its parts

3.1 Facts

Table	Key	Holds
`facts`	subject, family, `known_at`	every fact: `value` (jsonb), `status` (`ok`, `none`, `unavailable`), `reason`, `source`, `as_of`, `provisional`, `tool`
`facts_latest`	subject, family	the newest fact of each subject and family (the same columns)
`source_calls`	family, day	requests received, calls actually made, failures, the last error
`fact_parity`	day, family, subject, tool	while a family is in shadow: the store's answer against the tool's own, and the fields that differ

What is kept.

- **`facts_latest`:** every fetch updates it.
- **`facts`:** keeps every fetch, except for the fast families: the quote, the numbers derived from it, and the market context stamped from each state. Those enter the history only at discovery, at each tool's photo and at the close.

So the history stays small. Every photo holds its whole card anyway, and the market history is kept on its own.

The families, their writers and their clocks:

- the card's families: CARD.md;
- the O'Neil families: RESEARCH-WORKER.md.

3.2 The copy of the market history

The desk keeps the market history in its own file (MARKET.md). These are its copies, written at the same moment and compared every day, plus the comparison of the shadow weeks.

Table	Key	Holds
`market_states`	id	`at`, `day`, `status` (`ok`, `no_regime`), `reason`, `state` (jsonb): the indexes, the three reads, the regime, the 15 sectors, the hot sectors, the rules in use
`hot_sector_days`	sector, day	each sector's best score of the day, its counts toward hot and toward cooling, hot or not, hot since
`market_parity`	tool, `at`	while the market reader is in shadow: each tool's own read against the shared one

Detail: MARKET.md.

3.3 Research

Table	Key	Holds
`research_wanted`	ticker, day	each ticker a tool carded, with when and by which tool (written by the desk)
`research_attempts`	ticker, day	each attempt of the research worker for a ticker, ok or not, with detail
`research_runs`	id	each run: kind, start, end, ok, each step's result, counts

3.4 The register copies

Table	Key	Holds
`tools`	id	name, status, role, capture times and why, the tradability floor
`screeners`	tool, key	name, status, mirror, purpose, current version
`screener_versions`	tool, key, version	the definition (jsonb), its hash, when, a note
`r1_photos`	tool, day, ticker	`captured_at`, the card (jsonb, with the versions of the screeners that matched), `late`, `source`
`r3_outcomes`	tool, day, ticker, entry	entry price, highest high, lowest low, ATR14, upR, downR, opening range high and low, `captured_at`
`shortlist_items`	tool, day, ticker	when added, by which method, the item (jsonb), when exported

How they are written and checked:

- The desk writes each copy at the moment it writes the tool's own file.
- The two are compared every day.
- The old history of all eleven tools was imported once, with 0 differences.

qp reads them as of a day (BACKTEST.md).

3.5 Run records

Table	Holds
`desk_log`	notable events, such as a hot-sector reset
`discovery_scans`, `engine_runs`, `engine_answers`	the discovery engine's comparison record. The engine is off and retires with the move into the desk (ARCHITECTURE §10)
`schema_migrations`	the migrations that have run

3.6 Its switches

Each area has a switch in `data/keys.json`:

Setting	Means
off	not used: each tool works as before
shadow	written and compared beside the tool's own way; nothing depends on it
on	used

Switch	Area	Today
`factsMode`	short interest, borrow, news from the fact store	shadow
`marketMode`	the one market read	shadow

Switch	Area	Today
`researchMode`	the O'Neil facts on the cards	shadow
`registersMode`	the register copies	shadow: written and compared, and it stays there
`discoveryMode`	the discovery engine	off
`quantRegistersMode`	qp reading registers from here	off

A family or an area is switched **on** only when its comparison is clean.

- **For a fact family**, also when its call counts show the tools share enough tickers for sharing to pay (MIGRATION.md step D).
- **The register copies have no "on".** They stay copies, and each tool's own file stays the record (ARCHITECTURE §10). What is switched on is qp reading them (`quantRegistersMode`).

3.7 Where it lives

On Railway today: a Postgres reached over TLS, its address kept in `data/keys.json` as `databaseUrl`.

The other places it could live:

- on the box, at 60–100 MB of its 912 MiB;
- on AWS RDS.

This is an open decision for the trader (ARCHITECTURE §10). Moving it is one dump and one restore.

What guards it:

- **Saving the address.** It is checked before it is saved: it must be a Postgres address, reachable from the box (not Railway's private address), and it must connect.
- **Its own database only.** Migrations refuse a database that already holds another project's tables.
- **One TLS rule** for every program that connects.

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4. Contracts

Writer	Writes only
the desk	the card fact families, `source_calls`, `fact_parity`, the market tables, `research_wanted`, the register copies, `desk_log`, the discovery tables
the research worker	the O'Neil fact families, `research_attempts`, `research_runs`, the models (when chosen)
the backup	nothing: it reads, dumps and verifies

Reader	Reads
the desk	facts and O'Neil facts for its cards; the market state, only while the market history is not yet in its own file
qp	registers, market states and facts, as of a day
the journal	facts, the market state and R1 rows, as of a trade's entry

Migrations. They are numbered, run once and recorded, under an advisory lock so that two programs starting together cannot both migrate. Today they are `001_facts`, `002_market`, `003_research`,

`004_registers`, `005_discovery` and `006_engine_answers`, in `src/facts/index.js`.

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5. When it fails

Event	What happens
it is unreachable	every program stays off it for a minute, then tries again. Tools work from their own files and fetch their own facts. The copies and the shared facts wait, and the gap is shown on the health page
a write fails	the tool's own file already holds the row, so nothing is lost; the daily comparison finds the gap and the copy is written again
it is lost	restore last night's dump. The register copies can also be rebuilt from the tools' files (`scripts/import-registers.js`), with 0 differences proven once, and the market history from the desk's file
the address saved is wrong	refused before saving, with the reason (BOX.md §4.3)

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6. Built on

The old desk had no shared database:

- facts lived in each tool's memory, fetched once per tool;
- the O'Neil facts in JSON files written by qp;
- the market read was made five times a scan.

The design that made it: ARCHITECTURE §3.

The code:

- `src/facts/index.js` (the store, the migrations, the switches);
- `src/market/store.js`;
- `src/registers/store.js`;
- `quant-platform/research/worker.py`;
- the code repository's `docs/FACT-STORE.md` and `docs/REGISTERS-DB.md`.

What it learned:

- **Sharing a fact pays only when tools hold the same tickers.** The call counts decide each family.
- **The discovery engine saves nothing.** With every tool's screeners different, there is no identical scan to merge.

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7. Status

Part	Status
the store, its migrations, its switches, the TLS rule, the foreign-database guard	**built**, on Railway (2026-10-08)
facts, market states, research facts	**built, in shadow**. The market states are today the only store; as a copy of the desk's own history they come at the end of step B

Part	Status
register copies, with the history of all eleven tools imported (0 differences)	**built**, written and compared
each area switched on	**to do**, area by area, when its comparison is clean (step D)
`health_checks`, models	**to build**
where it lives	**open**: the trader's decision